## 5.2 THE OPEN ECONOMY IS-LM FRAMEWORK

After our discussion above on certain basic concepts and accounting tools pertaining to open economies (see Unit 4), we will now examine output market equilibrium in the short run Keynesian framework using the open economy IS-LM model.

As in case of the closed economy IS-LM model, here also we assume that prices are given and there is 'excess capacity', so that output is demand determined.

### 5.2.1 The IS Curve

You are familiar with the IS curve in a closed economy (see Unit 3). In an open economy, aggregate demand (AD) is given by:

$$AD = C + I + G + X - M = C + I + G + NX, \qquad (5.1)$$

where, $C = C(Y_d)$, is aggregate consumption expenditure which is a function of disposable income, $Y_d = Y - T$ ;

$I [= I(i)]$ is private investments, an inverse function of the rate of interest i;

G represents exogenous government expenditure;

$X [= X(Y_f, r)]$ is exports, the demand for domestic goods by foreigners, which is a function of foreign GDP ($Y_f$) and the real exchange rate r;

$M [= M(Y, r)[$ is imports, the demand for foreign goods by domestic agents, which is a function of domestic GDP (Y) and the real exchange rate r;

$NX [= (X - M) = NX(Y_f, Y, r)]$ stands for the net exports that depend on the factors underlying demand for imports and exports, i.e., $Y_f$, Y and r.

Clearly imports are a leakage from the domestic economy as it is the expenditure ofdomestic agents on output that is produced in a different economy (hence part of their GDP) while exports are an addition to the total expenditure on domestic goods. Exports depend on $Y_f$, *ceteris paribus*, the higher is $Y_f$, higher would be export demand. Similarly, the higher is Y, higher would be the demand for imports. Both exports and imports depend on the real exchange rate which represents the relative price of foreign vis-à-vis domestic goods as discussed in equation (6). With an increase in r (a real depreciation), foreign goods become relatively more expensive. Thus, there is a decline in imports, while exports increase as domestic goods become relatively cheaper. So a rise in r would lead to a rise in X and a fall in M and net exports would be higher. Note that economists Alfred Marshall and Abba Lerner gave certain conditions under which a real devaluation or depreciation leads to an improvement in the trade balance (increase in NX), known as the *Marshall- Lerner condition*. According to the Marshall-Lerner condition, depreciation will lead to an improvement in the balance of trade of a country only if the sum of the price-elasticities of exports and imports of the country is greater than one. Throughout our discussion we assume that the Marshall-Lerner condition is satisfied.

Note that one of the assumptions in the IS-LM framework is that of given prices. Therefore, with given foreign ($p^f$) and domestic prices ($p^d$), a nominal depreciation or appreciation (rise or fall in e) brings about a real depreciation or appreciation (rise or fall in r).

As in the closed economy case, the IS curve, represents goods market equilibrium, so that along the IS curve we have aggregate demand equal to aggregate output or GDP (Y):

i.e., Y = AD

or, $Y = C(Y_d) + I(i) + G + NX(Y_f, Y, r)$ (5.2)

However, compared to the closed economy case, now there are two new determinants of aggregate demand, viz., foreign GDP and the real exchange rate. These two factors affect net exports and hence demand and output in an open economy. If $Y_f$ increases (decreases) or r increases (decreases), NX would also increase (decrease) and the IS curve would shift to the right (left).

Another aspect of open economies reflected in (5.2) is that any change in aggregate demand (e.g., due to an increase in G), would affect Y and hence M and therefore lead to a change in NX. For instance, *ceteris paribus* an increase in government spending would lead to an increase in Y and M and thus reduce NX (i.e. reduce the size of the trade surplus or widen an existing trade deficit). However, if foreign GDP $Y_f$ increases, *ceteris paribus*, exports would increase (as these are imports of the foreign country), also leading to an increase in Y, but in this case the trade balance would improve, i.e., NX would rise (assuming that rise in X due to increase in $Y_f$, would be more than the rise in M induced by rise in Y).

### 5.2.2 The LM Curve

So far as the LM curve is concerned, openness of an economy does not fundamentally change the inverse relation between money demand and the rate of interest (see Unit 3). If the demand for money is higher, the lower is the rate of interest, as in the case of a closed economy. Thus the LM curve, representing money market equilibrium (equality between demand for money and supply of money) is upward sloping. However, in an open economy, asset transactions with the rest of the world would give rise to capital flows (recorded in the capital account of the BOP). This also affects the rate of interest and money supply in an open economy. We discuss further on this issue below.

Let us assume that there is ***perfect capital mobility***, i.e., there are no restrictions on international capital inflows and outflows and cross-border capital movements involve zero transaction costs. This is a simplifying assumption that captures the reality of large-scale international capital mobility observed in the era of globalization, with advances in technology greatly reducing the cost of moving capital across international borders.

The assumption of perfect capital mobility has certain implications for the determination of interest rates in an open economy. It means that the domestic and foreign bonds are perfect substitutes, so that therate of intereston domestic bonds (i)has to be in line withthe rate of interest on foreign bonds ($i_f$), i.e., $i = i_f$ must hold. Here we assume that the foreign interest rate is given and not influenced by domestic economic policy.

If $i_f$ is higher than i (i.e., $i_f > i$), foreign bondswould be preferred by investors owing to relatively higher returns on them. In this case, the domestic economy would experience capital outflows, as agents sell domestic bonds and buy dollars to purchase foreign bonds. The outflows continue until i rises (as price of domestic bonds fall) and equality is restored between i and $i_f$. When $i = i_f$, economic agents (such as households and firms) are indifferent between domestic bonds and foreign bonds.

Another point you should note regarding the money market in open economies relating to the supply of money and the role of foreign exchange reserves. Note that the stock of foreign exchange reserves is an asset of the central bank. When the central bank buys foreign currency, adding to its assets, it issues domestic currency so there is matching rise in its liability. As such, any change in foreign exchange reserves affects the monetary base or high powered money (liabilities of the central bank) and hence money supply in the economy, via the money multiplier.

Therefore, when a country experiences capital inflows and has a BOP surplus, this increasesforeign exchange reserves and leads to an increase in money supply. Similarly, a capital outflow can result in a BOP deficit and a fall in money supply via a reduction in the stock of foreign exchange reserves.

In this context you should note that the central bank often uses a policy of ***sterilization*** to offset the impact of a change in foreign exchange reserves on the money supply. This can be done by using open market operations. In particular, central banks often use foreign exchange reserves for intervention in the foreign exchange market. Suppose the RBI intervenes to buy dollars (in order to absorb an excess supply of dollars) against rupees. This would add to its foreign exchange reserves and hence increase money supply. However, the RBI can carry out a sterilized intervention, by simultaneously selling bonds, in an open market operation, to reduce money supply. Thereby the RBI can either fully or partially